The New CMOS GOA Circuit with Partitioned Frequency Division Function in AMOLED

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Abstract

8T2C LTPO Pixel technology has gained increasing recognition due to its low-frequency scenarios. However, the increase of GOA types will lead to an increase in IC power consumption, especially at 120 Hz. We propose a CMOS partitioned frequency division GOA circuit, which combines the GOA outputs of Nscan and Pscan to reduce IC power consumption by 8.36%~9.81% and reduce left and right borders by 75 μm~100 μm.

Author Keywords

8T LTPO pixel; CMOS GOA; IC power; border; partitioned frequency division.

1. Introduction

LTPO (low temperature polycrystalline oxide) technology, which amalgamates the attribution of LTPS and IGZO (indium gallium zinc oxide), has garnered substantial approbation from consumers within the fields of smartphones, watches and laptops [1]. The energy conservation mechanism of LTPO is anchored in the augmentation of low-frequency display applications. Notably, the 8T2C LTPO pixel circuit has received considerable favor among technicians, owing to its lower frequency, expanded scope of low-frequency applications and enhanced ghosting performance. Nonetheless, it is imperative that the 8T (TFT, thin film transistor) LTPO pixel circuit necessitates 5 types of GOA (gate on array) signal, which has escalated from 3 types in the preceding 7T LTPO pixel circuit, as depicted in **Figure 1** [2-7]. The increment in the variety of GOA types is poised to elevate IC (integrated circuit) power consumption, contravening the primary objective of the LTPO technology development [8-11]. Consequently, diminishing the IC power consumption of LTPO technology is an eternal pursuit and trend [12-15]. In the preceding year, we pioneered a technology pertaining to frequency division [16]. Building upon this foundation, we sophisticated a CMOS (complementary metal oxide semiconductor) GOA circuit integrated with a frequency division function. This innovation is geared towards achieving a reduction in the power consumption of LTPO ICs.

In this work, we have successfully established the CMOS GOA architecture, which concurrently encompasses the functionalities of both Nscan GOA and Pscan GOA typically found in the traditional LTPO architecture. The output Pscan signal waveform from the new CMOS GOA is identical to the Nscan signal waveform. Moreover, the border width could be significant reduced by consolidating the two circuits into one unit. Additionally, the IC power consumption has been reduced by 8.36% in the full-screen 120 Hz mode, by 9.81% in the zone-difference screen mode, and by 9.29% in the full-screen 1 Hz mode.

2. Results and Discussion

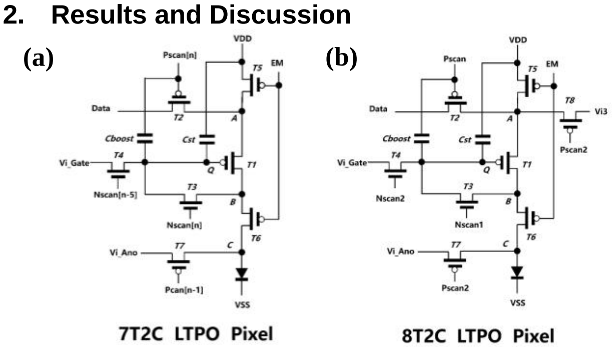

Figure 1. The schematic diagram of (a) 7T LTPO pixel circuit and (b) 8T2C LTPO pixel circuit

CMOS GOA circuit is introduced to improve the driving architecture of the pristine LTPO technology. In conventional panel design framework, as depicted in **Figure 2(a)**, there are three sets of GOA units on either side of the AA region, designated as EM/Nscan2/Pscan and Pscan/Nscan1/Pscan2. Typically, EM/Pscan2/Nscan1/Nscan2 serve as single-sided drivers, activating adjacent pixel rows, whereas Pscan operates as a double-sided driver, affecting a single pixel row. In CMOS GOA driving framework which consolidats Nscan and Pscan into a unified unit, one Nscan1 (or Nscan2) signal and two scan signals are output by the CMOS GOA, as illustrated in **Figure 2(b)**. EM/Pscan2 remain

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unchanged, and the Nscan signal output by the CMOS continues to be single-sided driven. The double-sided driving signal is composed of the Pscan signals from both flanks.

As shown in **Figure 3**, the CMOS GOA circuit unit consists of 25T4C (19 p-type TFTs, 6 n-type TFTs and 4 capacitors) components, where T9 and T10 buffer units are tasked with generating the Nscan (Nout) signal, while T6 and T7 buffer units produce the Pscan (Pout) signal. T24 and T25 function equivalently to the T6/T7 pair. In contrast to traditional circuit units, the Out signal exclusively drives pixels within the AA zone, with the level-shifting signal corresponding to the internal node signal P[n] within the midpoint of GOA, which exhibits a waveform similar to that of the Nout output.

This circuit contains two clock signals, where XCK (clock) and CKI/CK2 generats the Nout signal and the Pscan signal, respectively. The two clock signals do not interfere with each other and have no restrictions. The NVGH inside the GOA is responsible for generating the high level of the Nout signal, while NVGL is mainly responsible for generating the low level of the Nout signal and also responsible for the input of T13 and T14. PVGH maintains the stability of the internal nodes of the GOA and outputs the majority of the high level of the Pout signal.

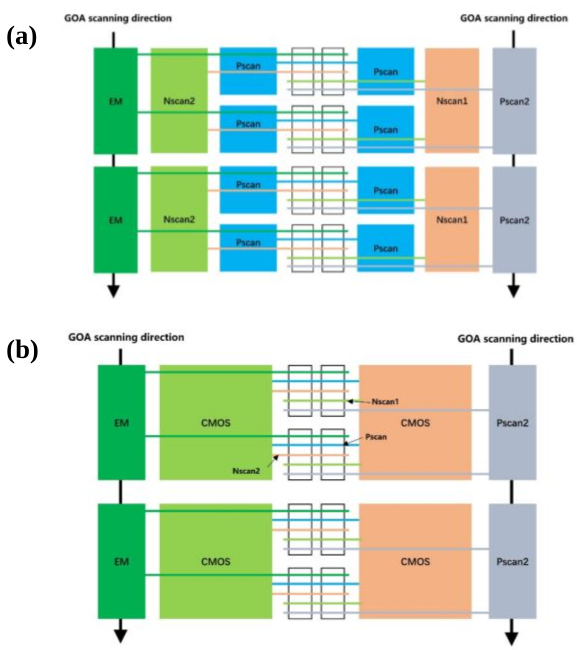

Figure 2. The schematic diagram of (a) conventional driving framework and (b) CMOS GOA driving framework

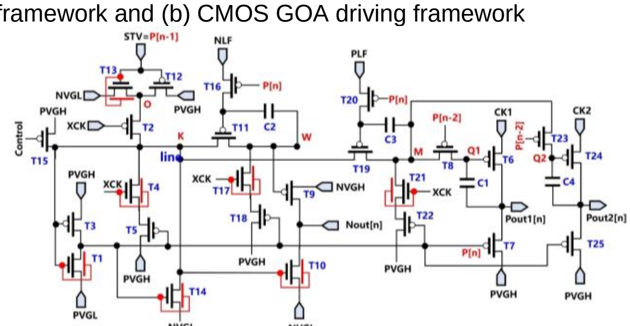

Figure 3. The schematic diagram of CMOS GOA circuit

The improved circuit also supports frequency division, in which the frequency division functions for Nscan and Pscan could be independently enabled. Specifically, T16/T11/T17/T18/C2 facilitate the enable of the Nout frequency division unit, with NLF acting as the critical signal for controlling the frequency division function of Nout. T19/T20/T21/T22/C3 enable the Pout frequency division unit, with PLF acting as the critical signal for controlling the frequency division function of Pout. The frequency division function remains inactive when NLF/PLF is at low voltage VGL, and when NLE/PLF is at high voltage VGH, the frequency division function is activated, as depicted in **Figure 4**.

For the unique framewok of CMOS GOA, Nout and Pout signals are generated by a group of GOA units, ensuring the position relationship between Nout and Pout signals is relatively fixed. Therefore, to synchronize with 8T LTPO pixel timing, the GOA to AA connection relationship for driving a row of pixels needs to be adjusted by invoking different rows of Nout and Pout inputs. For instance, as shown in **Figure 5**, the Nscan2 signal for driving T4 comes from the Nscan2 signal of the GOA unit on the left in the n-3th row, while the Nscan1 signal for driving T3 comes from the

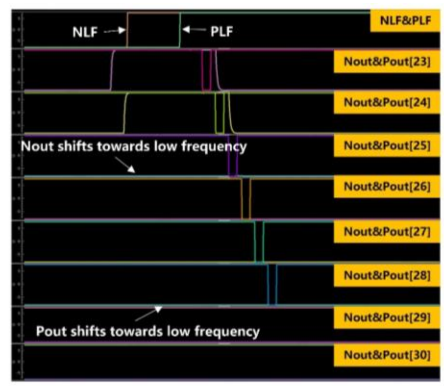

Figure 4. The waveform diagram of CMOS GOA circuit

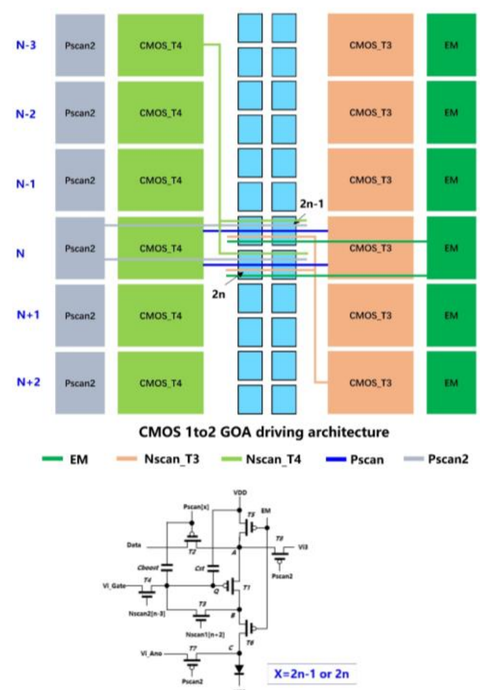

Figure 5. The schematic diagram of CMOS 1to2 GOA driving framework

Nscan1 signal of the GOA unit situated to the right in the n+2th row, and the corresponding Pscan signals are drived from the Pout1 and Pout2 outputs of the nth row GOA unit.

Figure 6. The physical photos of CMOS GOA panels with low power consumption

The modified CMOS GOA demonstrates a marked superiority relative to the traditional GOA, particularly with regard to the minimization of frame size and the reduction in IC power consumption (as shown in **Figure 6)**. Firstly, by merging the Nscan and Pscan GOA units into a single CMOS GOA unit, we could save space occupied by the Nscan and Pscan GOA units. Compared to our previous generation LTPO differential frequency GOA unit, the GOA width is reduced by ~75 μm, which facilitates the realization of narrower border design. Moreover, if the differential frequency function is disabled, the border can be further reduced by ~25 μm. Secondly, as shown in **Table 1**, we compared the power consumption data of two panel modes with different GOA circuits. Compared to the previous-generation LTPO differential frequency GOA circuit, after turning off all IC compensation IP, the sample drived by CMOS GOA reduces the IC power consumption by 8.36% in full-screen 120 Hz mode and 9.81% in differential frequency mode, and 9.29% in full-screen 1 Hz mode. The power savings of the IC mainly come from the savings of the Nscan CK signal and the VGH of the clock signal in low-frequency mode.

Table 1. IC power consumption of Normal GOA and CMOS GOA in defferent frequency modes

IC power consumption (mW)	Full screen (120 HZ)	Zone difference screen (10/60/10 HZ)	Full screen (1 HZ)
Normal GOA	242.66	142.23	94.13
CMOS GOA	219.26	125.07	82.37

3. Conclusions

In summary, we have modified the GOA architecture of traditional LTPO technology by incorporating IGZO into the GOA domain. By amalgamating the Nscan and Pscan, the issue of signal synchronization among disparate GOA output clusters within a pixel is effectively addressed. In comparison to the preceding generation of LTPO, we have realized a reduction in the lateral GOA dimensions by 75-100 μm, concurrently diminishing the IC power consumption by 8.36% in the full screen 120 Hz operation mode, by 9.81% in the differential mode, and by 9.29% in the full screen 1 Hz operation mode.

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